Hence, the solution is $50-T=-20e^{\frac{1}{2}t\log\left(\frac{1}{2}\right)}$ or $T=50+20\left(\frac{1}{2}\right)^{\frac{t}{2}}$

Now, we would like to find the value of t, for which $T(t)=98.6$, and $t=2\left(\frac{\log\left(\frac{48.6}{20}\right)}{\log\left(\frac{1}{2}\right)}\right)\approx -2.56$

It appears that the person was murdered at about 5.30 p.m.

10.8.4 Mixture problems

Mixing problems occur quite frequently in chemical industry. Now we explain how to solve the basic model involving a single tank.



Fig. 10.2

A substance S is allowed to flow into a certain mixture in a container at a constant rate, and the mixture is kept uniform by stirring. Further, in one such situation, this uniform mixture simultaneously flows out of the container at another rate. Now we seek to determine the quantity of the substance S present in the mixture at time t.

Letting x to denote the amount of S present at time t and the derivative $\frac{dx}{dt}$ to denote the rate of change of x with respect to t. If *IN* denotes the rate at which S enters the mixture and *OUT* denotes the rate at which it leaves, then we have the equation $\frac{dx}{dt}=IN-OUT$

Example 10.30

A tank contains 1000 litres of water in which 100 grams of salt is dissolved. Brine (*Brine is a high-concentration solution of salt (usually sodium chloride) in water*) runs in a rate of 10 litres per minute, and each litre contains 5grams of dissolved salt. The mixture of the tank is kept uniform by stirring. Brine runs out at 10 litres per minute. Find the amount of salt at any time t.

Solution

Let $x(t)$ denote the amount of salt in the tank at time t. Its rate of change is

$\frac{dx}{dt}=$ in flow rate $-$ out flow rate

Now, 5 grams times 10 litres gives an inflow of 50 grams of salt. Also, the out flow of brine is 10 litres per minute. This is $10/1000=0.01$ of the total brine content in the tank. Hence, the outflow of salt is 0.01 times $x(t)$, that is $0.01x(t)$.

Thus the differential equation for the model is $\frac{dx}{dt}=50-0.01x=-0.01(x-5000)$

This can be written as $\frac{dx}{x-5000}=-(0.01)dt$

Integrating both sides, we obtain $\log|x-5000|=-0.01t+\log C$

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